

## CSE205:DATA STRUCTURES AND ALGORITHMS

L:3 T:0 P:2 Credits:4

**Course Outcomes:** Through this course students should be able to

CO1 :: understand the time and space complexity of programs and data-structures.

CO2 :: illustrate the importance of Linked List in context of real world problems

CO3 :: differentiate the Stack and Queue data structures for problem solving

CO4 :: use of recursion in iteration process and tree data structure

CO5 :: apply heap operations and hashing techniques for efficient data storage, retrieval, and collision resolution.

CO6 :: identify and integrating emerging data structures and algorithmic ensuring their ability to stay with technological advancements and maintain professional growth in a dynamic field

### Unit I

**Introduction** : Basic Concepts and Notations, Complexity analysis: time space and trade off, Omega Notation, Theta Notation, Big O notation, Basic Data Structures.

**Arrays** : Linear arrays: memory representation, Array operations: traversal, insertion, deletion, sorting, searching and merging and their complexity analysis.

**Sorting and Searching** : Bubble sort, Insertion sort, Selection sort, Searching: Linear Search and Binary Search

### Unit II

**Linked Lists** : Introduction, Memory representation, Allocation, Traversal, Insertion, Deletion, Header linked lists: Grounded and Circular, Two-way lists: operations on two way linked lists

### Unit III

**Stacks** : Introduction: List and Array representations, Operations on stack (traversal, push and pop), Arithmetic expressions: polish notation, evaluation and transformation of expressions.

**Queue** : Array and list representation, operations (traversal, insertion and deletion), Priority Queues, Deques

### Unit IV

**Recursion and Trees** : Introduction to Recursion, Binary trees: introduction (complete and extended binary trees), memory representation (linked, sequential), Binary Search Tree: introduction, searching, insertion and deletion, In-order traversal, Pre-order traversal, Post-order traversal using recursion, Recursive implementation of Towers of Hanoi, Merge sort, Quick sort

### Unit V

**Heaps and Hashing** : Heaps: Introduction, Insertion, Deletion, HeapSort, Hashing introduction: hash functions, hash table, Open hashing (separate chaining), Closed hashing (open addressing): linear probing, quadratic probing and double hashing

### Unit VI

**Graphs** : Graph Traversal: BFS, DFS, Shortest path algorithm: Dijkstra's Algorithm, Bellman-Ford Algorithm, Floyd Warshall Algorithm

### List of Practicals / Experiments:

#### Array

- Program to implement insertion and deletion operations in arrays

#### Searching

- Program to implement different searching techniques - linear and binary search

#### Sorting

- Program to implement different sorting techniques – bubble, selection and insertion sort

#### Linked List

- Program to implement searching, insertion and deletion operations in linked list

**Doubly Linked List**

- Program to implement searching, insertion and deletion operations in doubly linked list

**Stack**

- Program to implement push and pop operations in stacks using both arrays and linked list

**Queues**

- Program to implement enqueue and dequeue operations in queues using both arrays and linked list

**Recursions**

- Program to demonstrate concept of recursions with problem of tower of Hanoi

**Recursive Sorting**

- Program to implement recursive sorting techniques - merge sort, quick sort

**Tree**

- Program to create and traverse a binary tree recursively

**Binary Search Tree**

- Program to implement insertion and deletion operations in BST

**Heaps**

- Program to implement insertion and deletion operations in Heaps and Heap Sort

**Text Books:**

1. DATA STRUCTURES by SEYMOUR LIPSCHUTZ, MCGRAW HILL EDUCATION

**References:**

1. DATA STRUCTURES AND ALGORITHMS by ALFRED V. AHO, JEFFREY D. ULLMAN AND JOHN E. HOPCROFT, PEARSON